



Forecasting the Consumer Price Index in Türkiye Using Machine Learning Models: A Comparative Analysis

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Highlights

- To forecast Türkiye's Consumer Price Index (CPI), machine learning models are employed.
- Data is used from 2012 to 2024 and the "SelectKBest" method identified 13 key indices.
- Comprehensive analyses are performed using multiple machine learning models.

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Abstract

This study utilizes machine learning models to forecast Türkiye's Consumer Price Index (CPI), thereby addressing a critical gap in inflation prediction methodologies. The central research problem involves the forecasting of CPI in a volatile economic environment, which is essential for informed policymaking. The primary objective of this study is to evaluate the performance of three machine learning models, such as Decision Tree (DT), Random Forest (RF), and Support Vector Machine (SVM), in forecasting CPI over periods ranging from one to six months, utilizing data from 2012 to 2024. The study's unique contribution lies in the application of the "SelectKBest" method, which identifies the most relevant indices, thereby enhancing the efficiency of the models. An ensemble method, Averaging Voting, is also employed to combine the strengths of these models, producing more accurate and robust predictions. The findings indicate that while the RF model consistently generates the most accurate forecasts across all shifts, the SVM model demonstrates a particular strength in the domain of short-term predictions. The ensemble model demonstrates a substantial performance improvement, with a R^2 value of 0.962 for one-month ahead of estimates and 0.956 for five-month forecasts. This combined approach has been shown to outperform individual models, offering a more reliable framework for CPI forecasting. The findings offer valuable insights for economic policymakers, enabling more precise and stable inflation predictions in Türkiye.

1. INTRODUCTION

In today's high-inflation environment, the development of expectations concerning inflation and wages has become a central topic among policymakers and scholars. Following the COVID-19 pandemic, the influence of inflation expectations on wage-setting processes has become even more pronounced due to rising inflation [1]. The increase in the cost of living within a community over a given period is referred to as inflation. While high inflation results in a decline in social wealth, low inflation hinders the accumulation of social wealth. Consequently, a community requires a certain degree of inflation to facilitate the growth of social wealth [2]. Various indexes are used to measure the negative impacts of extraordinary events like wars, economic crises, natural disasters, and pandemics. These indexes are developed by analyzing data that is routinely gathered from the citizens of the affected countries [3]. The Consumer Price Index (CPI) provides insight into the changing prices of a representative basket of goods and services. This allows for a clear understanding of inflationary trends and their impact on several types of goods and services [4]. CPI is a statistical measure that quantifies the rate of change in the prices of services and goods purchased by households. It represents the variation in average prices over a defined period. The CPI serves as an indicator of fluctuations in consumer prices for purchased products. By comparing a baseline period, which is typically assigned a value of 100, with the current period's consumer prices, changes can be assessed. The CPI is the most prevalent metric for monitoring inflation and the cost of goods and services acquired

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by the average consumer on a fixed basis [5]. The objective of the CPI is to achieve a balance between the accuracy of the price data it collects, and the costs associated with this data. To obtain reliable figures, it is necessary to take account of the costs of processing and storing the data, as well as the potential for inaccuracy in the case of volatile prices [6]. CPI is an invaluable tool for gauging the extent of inflationary or deflationary pressure, a phenomenon that has a direct impact on the lives of ordinary citizens and informs the trajectory of national macroeconomic policymaking. Despite its prominence, the statistical principles and influencing factors behind the CPI are often overlooked in economic analysis [7].

In contemporary high-inflationary environments, the forecasting of inflation and the identification of its driving factors have become pivotal subjects for policymakers and researchers. Türkiye, like many countries, faces significant challenges in predicting the Consumer Price Index, which directly influences economic decision-making and social welfare. Accurate CPI forecasting is imperative for effective economic planning, impacting such pivotal areas as monetary policy, wage negotiations, and the cost of living. This study aims to address the limitations of current CPI forecasting methods by employing machine learning techniques, namely Decision Tree (DT), Random Forest (RF), and Support Vector Machine (SVM) models, to predict Türkiye's CPI over various time horizons.

A distinctive feature of this study is the employment of the "SelectKBest" method for feature selection, which identifies the most influential indices affecting CPI. Furthermore, an ensemble method known as Averaging Voting is employed to enhance the accuracy of predictions. This technique combines the strengths of different models, such as Decision Tree, Random Forest, and Support Vector Machine, ensuring more robust and reliable forecasts. The integration of diverse models produces more stable and error-tolerant results, especially given the complexity and variability of economic data. The findings emphasize the potential of integrating machine learning methodologies with feature selection and ensemble techniques to furnish more reliable CPI forecasts, offering valuable insights for policymakers in Türkiye.

The rest of this study is organized as follows: firstly, in the "Literature Review," the literature on consumer price index estimation in various countries will be examined. Then, the details of the data and methodology used for the estimation of the Consumer Price Index in Türkiye will be explained. Moreover, 13 key indices determined using the "SelectKBest" method and machine learning models (DT, RF, and SVM) will be emphasized. Finally, the performance of these models will be evaluated, the results will be discussed, and implications for policymakers will be presented in the last part.

2. LITERATURE REVIEW

Numerous studies have attempted to forecast the CPI using various methodologies, each with varying degrees of success. Traditional models like ARIMA are commonly used for time series forecasting due to their simplicity and ability to model linear trends. However, ARIMA models often struggle with capturing non-linear patterns and fail to adapt quickly to sudden shifts in the economy. For example, ARIMA-based forecasts have been widely used in countries such as, Ukraine [8], Uganda [9], Germany [10], Hungary [11], Nigeria [12], Türkiye [13], Malaysia [14], Ghana [15], Rwanda [16], Bangladesh [17], Somaliland [18], Belgium [19], Indonesia [20], China [21] and India [22]. While these models are effective for short-term forecasts under stable economic conditions, their performance tends to degrade when the economy undergoes rapid changes or exhibits volatility. In recent years, machine learning techniques have gained popularity for CPI forecasting due to their ability to handle complex, non-linear relationships in data. Models like Long Short-Term Memory (LSTM) networks have been shown to outperform ARIMA for long-term forecasts by capturing long-range dependencies in data [22]. Furthermore, Random Forest (RF) has demonstrated superior performance, particularly when dealing with large datasets, as it is robust to overfitting and can model complex interactions between features. RF has been used in countries like Saudi Arabia [23] and the United Kingdom [24] to forecast economic indicators, including CPI. Support Vector Machines (SVM) have been applied in CPI forecasting due to their ability to handle both classification and regression tasks, particularly in situations involving non-linear data. SVMs have shown impressive performance for short-term forecasting of CPI, particularly when paired with kernel functions to capture

complex patterns in the data. SVM models have been successful in countries like Saudi Arabia [23], the United Kingdom [24], China [25], Ecuador [26], United States [27], where the economic landscape is complex and highly non-linear. However, SVMs can be sensitive to the choice of hyperparameters and may require careful tuning to optimize performance.

Beyond these methodologies, alternative models have been proposed within the existing body of literature. Among the most notable studies are those conducted in India, which utilize models such as XGBoost, Theta, Prophet [22]. In Saudi Arabia, a range of models, including Linear Regression (LR), k-nearest neighbor, and DT, are utilized [23]. Ridge, Lasso, and Elastic models are used predicted CPI in UK [24]. Moreover, Artificial Neural Network (ANN) is used for forecasting CPI for different countries [23, 24]. Advanced techniques like Ensemble Empirical Mode Decomposition (EEMD) are preferred for predicting CPI in China [25]. In addition, various methods such as Particle Filter (PF), Kalman Filter (KFF) are used in Ecuador [26]. In the United States, Multivariate Linear Regression (MLR), Autoregressive Distributed Lag (ARDL), and Multivariate Adaptive Regression Splines (MARS) have been adopted [27]. In Canada, Recurrent Neural Network (RNN) with Teacher Forcing and Attention Mechanism is used [28]. Furthermore, advanced techniques such as Genetic Programming (GP) are preferred for forecasting CPI [29]. The Holt-Winters model is employed in Kenya, South Africa, and Saudi Arabia for CPI forecasting [30, 31]. While the GSTARI model is extensively employed in Indonesia [32], In Ghana, Exponential Smoothing (ES) methods [33] are employed. Furthermore, in the United Kingdom multivariate singular spectrum analysis (MSSA(AI)) [34] are employed. Multilayer Perceptron (MLP) has been applied in Indonesia [35], besides Double Exponential Smoothing (DES) is used for forecasting CPI in Indonesia [36]. SARIMA and Error Trend and Seasonality (ETS) models also are used for predicting CPI in Albania [37].

This literature review provides a comprehensive overview of the current research trends in this field, presenting the diversity and application areas of the time series forecasting methods used for CPI forecasting on a global scale, and Table 1 shows the summary of studies on CPI forecasting. On the other hand, the Symbolic Aggregate Approximation (SAX) method is used in trend analysis of time series natural gas consumption data [38] and BIST100 stocks [39], Association Rules analysis is used for predicting Traffic Accidents [40] and ES, HLT, Holt-Winters Damped Trend, Moving Average and ARIMA is used for predicting production of grains [41]. Moreover, SVM, RF, Bagging, and DT are used for the classification of historical coins [42]. SVM and ANN is used for the unemployment rate in Türkiye [43].

There are several notable gaps in the extant literature on forecasting the CPI. Primarily, traditional time series models and econometric methods, such as ARIMA and SARIMA, frequently prove incapable of efficaciously handling non-linear relationships, multicollinearity, and high-dimensional data within economic datasets. While machine learning techniques have seen increased utilization in recent years, their employment in CPI forecasting remains limited, and the potential of these models to capture complex economic relationships has not been fully explored. This study aims to address this gap by employing advanced machine learning models, including DT, RF, and SVM, to better capture both linear and non-linear patterns in the data. A further key gap in the literature is the insufficient integration of feature selection methods in CPI forecasting. A significant number of studies have employed the full range of economic indices without adequately identifying the most influential variables, resulting in inefficient models that are prone to overfitting or underfitting. To address this limitation, this study proposes the implementation of the "SelectKBest" feature selection method, which identifies the most relevant and predictive variables. This approach is expected to enhance the efficiency and performance of the models. In addition, although ensemble methods are effective, their utilization in CPI forecasting is limited. This study employs the Averaging Voting ensemble method, combining the strengths of individual models (DT, RF, and SVM) to provide more robust and accurate forecasts. Moreover, while most studies concentrate on short-term CPI forecasting, this research extends the scope by forecasting CPI across multiple time horizons, ranging from one to six months, thus offering a more comprehensive approach to inflation

forecasting. Finally, most of the extant research has been conducted in more stable economic environments, whereas this study focuses on Türkiye, a country with a dynamic and often volatile economy.

Table 1. Summary of studies on consumer price index (CPI) forecasting

Paper	Country	Methodology
[8]	Ukraine	ARIMA Models
[9]	Uganda	ARIMA Models
[10]	Germany	ARIMA Models
[11]	Hungary	ARIMA Models
[12]	Nigeria	ARIMA Models
[13]	Türkiye	ARIMA Models
[14]	Malaysia	ARIMA, Vector Autoregression (VAR) Models
[15]	Ghana	ARIMA, SARIMA Models
[16]	Rwanda	ARIMA Models
[17]	Bangladesh	ARIMA Models
[18]	Somaliland	ARIMA Models
[19]	Belgium	ARIMA Models
[20]	Indonesia	ARIMA models, ANN
[21]	China	ARIMA Models
[22]	India	XGBoost, Theta, ARIMA, Prophet, LSTM
[23]	Saudi Arabia	LR, RF, SVM, k-Nearest Neighbors (k-NN), DT, ANN
[24]	United Kingdom	Ridge, Lasso, Elastic, SVM, RF, ANN
[25]	China	EEMD, SVM
[26]	Ecuador	PF, SVM, KFF, ES, 4Theta, Fast Fourier Transform (FFT)
[27]	United States	MLR, SVM, ARDL, MARS
[28]	Canada	RNN with Attention Mechanism and Teacher Forcing
[29]	USA	ANN, GP, SARIMA
[30]	Kenya, South Africa	Holt Winter's Model
[31]	Saudi Arabia	HLT
[32]	Indonesia	GSTARI Model
[33]	Ghana	ES
[34]	United Kingdom	MSSA(AI)
[35]	Indonesia	MLP, LSTM
[36]	Indonesia	DES
[37]	Albania	ETS, SARIMA Models

3. MATERIAL AND METHOD

In the literature, time series analysis methods, machine learning methods, and ANN have been used in the forecast of the consumer price index. In this study, the indices affecting the CPI values are determined with the "SelectKBest" method, and the CPI values shifted from 1 to 6 months are estimated with the DT, RF, and SVM methods. Comparing the performances of the models, R^2 , Mean Absolute Percentage Error', Mean Absolute Error', Mean Squared Error', and 'Adjusted R^2 ' values are calculated. Figure 1 shows the summary of the methodology of this research.

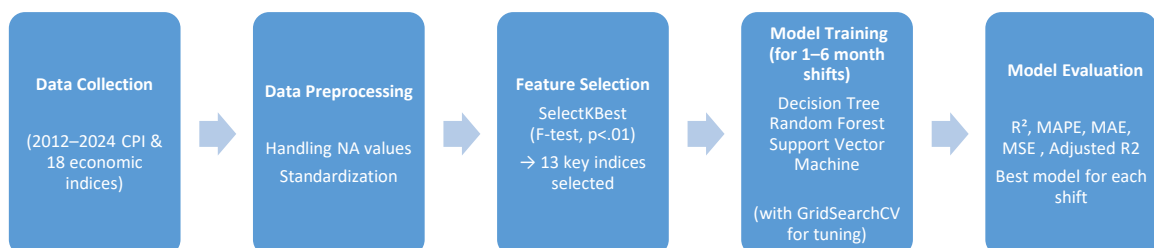


Figure 1. Summary of the methodology of this research

3.1. Data Pre-Processing

For the analysis, 18 different indexes announced by the Turkish Statistical Institute are used to predict CPI. These indexes are “Financial situation of household at present compared to the past 12 months”, “Financial situation expectation of household over next 12 months”, “General economic situation at present compared to the past 12 months”, “General economic situation expectation over next 12 months”, “Number of people unemployed expectation over next 12 months”, “Assessment on spending money on semi-durable goods over next 3 months compared to the past 3 months”, “Buying time condition of durable goods at present”, “Assessment on spending money on durable goods over next 12 months compared to the past 12 months”, “Saving time condition at present”, “Statement on current financial situation of household”, “The probability of saving over next 12 months”, “The probability of borrowing money over next 3 months”, “Assessment on consumer prices change rate over last 12 months”, “Expectation for consumer prices change rate over next 12 months compared to the past 12 months”, “Expectation for wages change rate over next 12 months compared to the past 12 months”, “The probability of buying a car over next 12 months”, “The probability of spending money on home improvements or renovations over next 12 months”, “The probability of buying or building a home over next 12 months”.

Feature selection in this study is conducted using the "SelectKBest" method from Python's "sklearn" library. SelectKBest is a filter-based technique commonly used in machine learning for selecting important features. It evaluates and ranks features by applying various statistical tests, such as the chi-squared test, ANOVA F-test, or mutual information score, to measure their relevance to the target variable. The method then selects the top K features with the highest scores, which are included in the final feature set. The scoring function computes the F-value between each feature and the target variable, where a higher score indicates a stronger association between the feature and the target [44]. In this study, analyzing F-test is used, and 0.01 is selected as the confidence interval. Indices with p-values less than 0.01 are used for model training. In addition to this, p-values of “Number of people unemployed expectation over next 12 months”, “Saving time condition at present”, “Statement on the current financial situation of household”, and “The probability of spending money on home improvements or renovations over next 12 months” and “The probability of buying or building a home over next 12 months” are 0.2975, 0.9722, 0.1514, 0.0151 and 0.8474 respectively. Moreover, they are extracted from the training dataset of the models because of high p-values according to our confidence interval. Standardization is a process that aims to ensure the accuracy and reliability of data by scaling it to a standard normal distribution, thus making it appropriate for machine learning models. Addressing potential issues such as missing data and scaling, the data preprocessing phase establishes the foundation for robust model training

$$\frac{X - \mu}{\sigma} = X_{normalized} \quad (1)$$

Standardization is applied to the remaining 13 features, and the analysis of the models is started. The data is scaled for a standard normal distribution, which is characterized by a mean of zero and a standard deviation of 1. The mean and standard deviation of each feature are calculated separately. The normalized calculation of the features is calculated via Equation (1). The standardization process is repeated for each shifted CPI value and the "NA" values are deleted.

3.2. Model Training

Forecasting will be made with the preprocessed data. Decision Tree, Random Forest, and Support Vector Machines are used as forecasting models. Decision Trees are frequently employed in addressing regression problems. By specifying a node's predicted value as a constant prediction and defining the node's impurity as the sum of squared deviations from the meaning of its samples, a decision tree can effectively function as a regression model [45]. Furthermore, the Random Forest algorithm, which is an ensemble method combining multiple decision trees, operates based on two key principles: bagging and random subspace methods. This approach is widely used for classification, regression, and clustering tasks. The primary concept involves training many decision trees on randomly selected subsets of the data, which collectively

produce highly accurate predictions by mitigating overfitting [46]. Support Vector Regression (SVR), a variant of Support Vector Machines, is applied for both linear and non-linear regression tasks using appropriate kernel functions [47].

Models are trained for each shifted CPI value. 0.8 of the data is used for training and 0.2 for testing the models. Parameter optimization is performed for using GridSearch for each shifted data set and fold. Parameter optimization involves selecting the best set of hyperparameters for a learning algorithm. A hyperparameter is a variable that influences the learning process. Hyperparameter optimization is a procedure aimed at finding a set of hyperparameters that produce the best model, which is defined as minimizing a specified loss function on provided independent data [48]. The combinations that gave the highest "R2" value are selected for each shifted data. The parameters used for GridSearch for each model are listed in Table 2.

Table 2. The parameters used for GridSearch for each model

Decision Tree	Max Depth	(0-100) increasing by 10
	Min Samples Split	(0-50) increasing by 5
	Min Samples Leaf	(1-20) increasing by 1
Random Forest	n Estimators	[100, 200, 300]
	Max Features	['sqrt', 'log2']
	Max Depth	[None, 10, 20, 30]
	Min Samples Split	[2, 5, 10]
	Min Samples Leaf	[1, 2, 4]
	Bootstrap	[False, True]
Support Vector Machine	C	(100-900) increasing by 10
	Epsilon	(10-300) increasing by 10
	Kernel	[linear, rbf, poly]

The adjustment of hyperparameters is pursued with the objective of enhancing model performance, with a particular focus on the key performance indicators (KPIs) of R2, MAPE, MAE, MSE, and Adjusted R2 which are utilized for the evaluation of the model. Following the training of the models, their capacity to predict CPI values is evaluated across a range of time shifts (1 to 6 months).

3.3. Model Evaluation

The performance of the forecasting models was assessed using five Key Performance Indicators (KPIs), namely R2, Mean Absolute Percentage Error (MAPE), Mean Absolute Error (MAE), and Mean Squared Error (MSE). The R2 score measures the proportion of variance in the dependent variable that is predictable from the independent variables, with values closer to 1 indicating stronger model performance. MAPE, on the other hand, quantifies the average percentage error between the predicted and actual values, providing a relative measure of forecast accuracy. MAE represents the average magnitude of errors between predicted and actual values and is particularly effective for evaluating model performance in the presence of outliers, as it is less sensitive to extreme values. MSE, which calculates the average squared differences between predicted and actual values, is a more sensitive indicator to larger errors, as it penalizes larger deviations more heavily. Lower MSE values signify better model performance. 5 KPIs are calculated for each forecast model and each shifted data as shown in Table 3.

The results of the evaluation process serve to highlight the strengths and weaknesses of each model. The Random Forest model, for instance, demonstrates consistent efficacy across all forecast shifts, while the SVM model exhibits proficiency in short-term predictions but demonstrates variability in long-term forecasting. The ensemble method, Averaging Voting, has been demonstrated to enhance the forecasting accuracy by combining the strengths of the constituent models. By comparing these results, we can draw conclusions about which model or combination of models best forecasts CPI.

4. RESULTS AND DISCUSSION

The findings of the machine learning models illustrate the pivotal function of feature selection in enhancing the precision of CPI forecasting. The employment of the "SelectKBest" method led to the identification of 13 features that were deemed to be the most statistically significant for predicting CPI, including indices about consumer expectations, financial conditions, and macroeconomic indicators such as unemployment rates and consumer confidence. Of these features, it was determined that those about consumer financial expectations, notably those associated with the household financial situation and future consumer price expectations, were found to have the most significant influence on the forecasting of CPI. These indices, which are indicative of consumer sentiment and future price expectations, demonstrated a strong correlation with CPI values. Furthermore, features such as "Number of people unemployed expectation over next 12 months" were found to have a substantial effect, especially in long-term CPI predictions, as labor market conditions directly influence consumer spending and, consequently, inflation. Purchasing power and consumer confidence indicators were also identified as critical factors, as reflected in the analysis. These indicators, such as the probability of saving over the next 12 months and an assessment on spending money on semi-durable goods over the next 3 months, are particularly important, as they reflect a consumer's capacity to spend, which in turn impacts demand-driven inflation. The effective dimensionality reduction, achieved through the process of feature selection, enabled the models to concentrate on the most significant variables. This, in turn, resulted in an enhancement of model performance and a reduction in the occurrence of overfitting. As a result, the models, Table 3 compare the performance metrics for CPI forecasting using DT, RF and SVM models over a shifting period of 1 to 6 months. The DT model exhibited high predictive accuracy, particularly at the 2-month and 5-month shifts, with R2 values of 0.926 and 0.966, respectively. However, it showed greater variability across other shifts. The RF model demonstrated consistently strong performance across all shifts, with R2 values ranging from 0.883 to 0.953 and the lowest MAPE of 0.081 on the 6-month shift. The SVM model achieved the highest R2 value of 0.970 in the 1-month shift but showed a decline in performance for longer forecast horizons.

The findings indicate that the RF model consistently provided the most stable and accurate forecasts across all shifting periods, with R2 values ranging from 0.883 to 0.953. Furthermore, it exhibited the lowest MAPE of 0.081 in the 6-month shift, demonstrating its robustness in handling dataset variations and providing reliable long-term forecasts. The DT model demonstrated high predictive accuracy at specific intervals but was less consistent across other shifting periods, indicating a higher sensitivity to input feature changes and potential overfitting. The SVM model exhibited remarkable short-term prediction capability, attaining the highest R2 value of 0.970 at the 1-month shift. However, its performance declined for longer forecast horizons.

The RF model is identified as the most reliable and versatile for multi-month CPI forecasting, while the SVM model is particularly effective for short-term predictions. The DT model, despite its variability, offers valuable insights at certain intervals, making it a useful tool in a comprehensive forecasting strategy.

Table 3. Comparison table of forecasting models according to R2, MAPE, MAE, MSE

DECISION TREE					
Shifted	R2	MAPE	MAE	MSE	Adjusted R2
1 month	0.876	0.106	70.848	25739.392	0.861
2 months	0.926	0.095	61.947	14721.449	0.917
3 months	0.830	0.101	78.384	31145.914	0.808
4 months	0.330	0.130	102.263	66532.898	0.244
5 months	0.966*	0.055*	32.128*	3503.180*	0.961*
6 months	0.544	0.095	81.437	45944.824	0.484
RANDOM FOREST					
Shifted	R2	MAPE	MAE	MSE	Adjusted R2
1 month	0.953	0.083	50.770*	9768.422	0.947*
2 months	0.934*	0.092	58.391	13119.865	0.926

3 months	0.896	0.084	64.526	19035.692	0.883
4 months	0.918	0.091	53.581	8148.335*	0.907
5 months	0.907	0.095	57.421	9545.724	0.895
6 months	0.883	0.081*	53.853	11824.750	0.867
SVM					
Shifted	R2	MAPE	MAE	MSE	Adjusted R2
1 month	0.970*	0.110	48.895*	6165.007*	0.967*
2 months	0.894	0.147	80.405	21312.991	0.880
3 months	0.942	0.116	59.733	10533.372	0.935
4 months	0.646	0.196	100.292	35189.364	0.600
5 months	0.933	0.101*	49.942	6840.707	0.924
6 months	0.576	0.184	108.060	42661.396	0.521

* Best values of each KPI

On the other hand, new predictions are created using the ensemble method, which is Averaging Voting, to ensure that the models provide more accurate and robust results. Ensemble methods are techniques that aim to produce more robust and reliable estimates by combining estimates from multiple models. These methods improve forecasting performance by overcoming the limitations of individual models and provide more generalizable results. Studies show that ensemble methods generally outperform a single model and reduce the risk of overfitting. The potential of ensemble methods could increase overall accuracy and reduce estimation error [49]. In addition, the bagging and random forest algorithms offer significant advantages in terms of model diversity and error reduction [50]. Moreover, the Boosting algorithm is effective in increasing estimation accuracy through sequential training of weak models [51]. These studies show that the use of ensemble methods in the estimation of complex economic indicators such as the CPI can contribute to the development of more reliable economic analysis and policy by increasing estimation accuracy. In this study, the voting-averaging technique is used to forecast CPI values. A voting-averaging (VOA) method is an ensemble technique for regression tasks. VOA is capable of dynamically selecting ensemble components through the utilization of the latent selectivity mechanism inherent to the voting process. Consequently, it can be posited that VOA represents an enhancement and expansion of both the voting and averaging methodologies [52].

Table 4. R2, MAPE, MAE, and MSE values of combined predictions

COMBINED RESULTS					
Shifted	R2	MAPE	MAE	MSE	Adjusted R2
1 month	0.962*	0.088	45.961	7862.480	0.957*
2 months	0.941	0.097	58.862	11760.772	0.934
3 months	0.914	0.077	56.754	15646.472	0.904
4 months	0.748	0.115	67.566	25075.385	0.715
5 months	0.956	0.074*	39.716*	4542.719*	0.950
6 months	0.721	0.103	75.443	28047.600	0.685

* Best values of each KPI

Table 4 shows the different KPI values for combined predictions. In this study, the ensemble method, which is created by taking the arithmetic average of the estimates of SVM, DT and RF models, has demonstrated considerable success in estimating CPI. The ensemble method demonstrated proficiency in mid-term forecasts, specifically for time horizons between three and five months. This period is often challenging for individual models, as it necessitates the effective capture of both long-term trends and short-term fluctuations. The VOA method was able successfully to combine the robustness of the Random Forest in capturing long-term trends with the strength of the SVM model in short-term predictions. This combination resulted in more stable and reliable mid-term CPI forecasts.

Nevertheless, while the ensemble approach effectively mitigated biases present in individual models, it encountered constraints in fully leveraging the exceptional capabilities of specific models during forecast intervals. In the context of short-term predictions (1-2 months), the SVM model exhibited superior performance, attaining higher R^2 values, particularly in the initial month, with an R^2 of 0.970. This is attributable to the fact that SVM, under its use of the kernel trick, is highly effective in capturing short-term fluctuations in the CPI data. In contrast, for predictions over a longer timeframe (6 months), the Random Forest model exhibited superior performance in comparison to the VOA method. The Random Forest model consistently yielded stable and accurate forecasts, achieving R^2 values ranging from 0.883 to 0.953 across different shifts. In comparison, the VOA method exhibited slightly diminished performance during the 6-month forecast period.

Conclusively, the VOA method is shown to be an effective approach for mid-term CPI forecasting, where its effectiveness lies in combining the strengths of different models and mitigating their weaknesses. However, for short-term and long-term forecasts, SVM and RF, respectively, delivered superior results. The ensemble method's primary contribution is in providing a balanced and more reliable prediction for forecast periods that lie between these extremes.

Table 5 summarizes the strengths and weaknesses of each model, and the ensemble method used in this study. The DT model demonstrates notable efficacy in the 2–5-month forecasting period. It functions effectively with smaller datasets and effectively captures non-linear relationships; however, it is susceptible to overfitting and demonstrates variability in performance across different periods. The RF model, on the other hand, demonstrates high accuracy across all forecast periods (1-6 months) and effectively handles large datasets, mitigating the risk of overfitting and producing reliable results. Nevertheless, the Random Forest model is computationally expensive, necessitating greater time investment during the training phase. The SVM demonstrates strength in the domain of short-term predictions by effectively capturing complex non-linear relationships. However, its efficacy is diminished in the context of long-term forecasts, and it is susceptible to hyperparameter settings. The VOA ensemble method, on the other hand, has been demonstrated to be effective by balancing the weaknesses of individual models (3-5 months). While VOA enhances the robustness and accuracy of the model, it is less effective in the short-term and long-term predictions when compared to individual models.

Table 5. *The strengths and weaknesses of each model*

Model	Best Forecasting Period	Advantages	Disadvantages
Decision Tree	2-5 months	Easy to interpret; performs well on small datasets and captures non-linear relationships.	Sensitive to overfitting; variable performance across different periods.
Random Forest	All periods (1-6 months)	High accuracy; handles large datasets well; reduces overfitting.	Computationally expensive; it requires more time to train.
Support Vector Machine	1 month	Excels in short-term predictions; captures complex non-linear relationships.	Less effective for long-term forecasts; sensitive to hyperparameter settings.
Ensemble Method (VOA)	3-5 months	Balances the weaknesses of individual models; improves robustness and accuracy.	Less effective in short-term and long-term forecasts compared to individual models.

5. CONCLUSION

The objective of this study is to forecast the Consumer Price Index (CPI) in Türkiye using a variety of machine learning models, including Decision Tree (DT), Random Forest (RF), and Support Vector Machine (SVM), over a range of forecast horizons, from one to six months. By employing the "SelectKBest" method, we could identify the most significant indices that impact CPI and utilize these features to train our models. The results demonstrated the varied performance of the models, illustrating their respective strengths and

weaknesses in different forecasting contexts. The results demonstrate that the RF model is the most reliable and adaptable for multi-month CPI forecasting, whereas the SVM model demonstrates superior performance in short-term predictions. Despite its variability, the DT model offers valuable insights at certain intervals, thus making it a useful component in a comprehensive forecasting strategy.

This study makes a significant contribution to the field of CPI forecasting, particularly in the context of Türkiye, by addressing the unique challenges posed by the country's dynamic and volatile economic conditions. Previous studies, which rely heavily on conventional techniques such as ARIMA or SARIMA models, have proven incapable of adequately accounting for the recurrent fluctuations in Türkiye's inflation, currency value, and broader economic shifts. In contrast, this research utilizes a comprehensive dataset and sophisticated machine learning models, meticulously adapted to the economic specifics of the Turkish context. The incorporation of pertinent indicators, namely consumer sentiment, unemployment expectations, and purchasing power, ensures that the study provides a more comprehensive and accurate forecasting framework, with factors highly relevant to Türkiye's inflation trends. The employment of the "SelectKBest" approach for feature selection ensures that the models prioritize the identification of the most significant variables, while the ensemble method, Averaging Voting (VOA), augments model robustness and reliability. In models might better capture short-term volatility and long-term trends. This study provides significant insights for policymakers in Türkiye, offering them reliable tools for managing inflation and making informed economic decisions.

To enhance the accuracy and robustness of our predictions, an ensemble method is employed utilizing Averaging Voting, which combines the strengths of SVM, DT, and RF models. This approach led to a notable improvement in the forecast performance, particularly in key performance indicators such as R2, MAPE, MAE, and MSE. The ensemble model exhibited superior performance compared to the individual models, with an R2 value of 0.962 and MAPE value of 0.088 in one-month ahead estimates. Similarly, in five-month ahead estimates, the ensemble model achieved an R2 value of 0.956 and MAPE value of 0.074, thereby demonstrating its capacity to provide forecasts of greater precision and reliability.

In conclusion, this study proposes a suite of models and methods for CPI forecasting, with the unique economic context of Türkiye serving as the basis for their development. The principles underlying these models are adaptable to different countries and economic indicators. The machine learning models used Decision Tree, Random Forest, and Support Vector Machine are highly versatile, with a proven track record in the global forecasting of economic indicators, including inflation rates, GDP growth, and unemployment rates. It is important to note that the "SelectKBest" feature selection method and the ensemble approach (VOA) are not unique to Türkiye and can be applied to any dataset where feature selection and model robustness are of crucial importance. The extent to which findings might be generalized is dependent upon the availability and relevance of data in disparate geographical contexts. For instance, the consumer sentiment, unemployment expectations, and financial indicators utilized in this study are valuable predictors for CPI in other countries with similar economic structures. However, it is important to note that in countries characterized by markedly divergent economic conditions, a modification of the feature selection process might be necessitated to accommodate the distinctive economic indicators that are characteristic of these respective regions.

While the models and methodologies were developed with specific challenges in mind, the overarching approach is adaptable and has the potential to be applied in other contexts to test its effectiveness in different economic environments. This adaptability makes the proposed framework a valuable tool for broader economic analysis and forecasting. The ensemble method serves to enhance the predictive accuracy and robustness of the model, thereby making it a valuable tool for economic analysis and policymaking. Further research could investigate the application of these methods to other economic indicators and in different geographical contexts, to validate their generalizability and effectiveness.

DECLARATION OF ETHICAL STANDARDS

The authors of this article declare that the materials and methods used in this study do not require ethical committee permission and/or legal special permission.

CONFLICTS OF INTEREST

No conflict of interest was declared by the authors.

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